



Maintenance Schedule

Diesel Engine

8V396TE54

12V396TE54

16V396TE54

Application group 3A

M050703/03E

Engine model	kW/cyl.	Application group
8V396TE54	85 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 50 Hz
12V396TE54	86 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 50 Hz
16V396TE54	85 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 50 Hz
8V396TE54	99 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 60 Hz
12V396TE54	100 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 60 Hz
16V396TE54	99 kW/cyl.	3A, continuous operation, unrestricted, ICXN, 60 Hz

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Load profile

Load profile

Load	100%	50%	10%
Corresponding operating time	5%	80%	15%

1.4 Maintenance schedule matrix

0 to 7,250 Operating hours

Task	Limit	Daily	Operating hours [h]																												
			250	500	750	1,000	1,250	1,500	1,750	2,000	2,250	2,500	2,750	3,000	3,250	3,500	3,750	4,000	4,250	4,500	4,750	5,000	5,250	5,500	5,750	6,000	6,250	6,500	6,750	7,000	7,250
W1096	2 a																														
W0500	-	X																													
W0501	-	X																													
W0502	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0507	-	X																													
W0508	-	X																													
W1024	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1009	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1080	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1002	-					X				X				X				X				X				X				X	
W1027	1 a					X				X				X				X				X				X				X	
W1053	1 a					X				X				X				X				X				X				X	
W1001	2 a					X				X				X				X				X				X				X	
W1061	-									X								X								X					
W1060	-									X								X								X					
W1174	-									X								X								X					
W1011	-									X								X								X					
W1176	2 a									X								X								X					
W1005	2 a									X								X								X					
W1189	2 a									X								X								X					
W1038	-																														
W2000	-																														
W2002	-																														
W2003	-																														
W2009	-																														
W2010	-																														
W2011	-																														
W2017	-																														
W2036	-																														
W2041	-																														
W2081	-																														
W2110	-																														
W2111	-																														
W2112	-																														
W2062	-																														
W1063	-																														
W3000	18 a																														
W3001	18 a																														
W3003	18 a																														
W3004	18 a																														
w = weeks m = months a = years																															

w = weeks
m = months
a = years

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Task	Limit	Operating hours [h]																			
		Daily	250	500	750	1,000	1,250	1,500	1,750	2,000	2,250	2,500	2,750	3,000	3,250	3,500	3,750	4,000	4,250	4,500	4,750
W3005	18 a																				
W3009	18 a																				
W3015	18 a																				
W3017	18 a																				
W3018	18 a																				
W3019	18 a																				
W1043	18 a																				
w = weeks m = months a = years																					

7,500 to 14,750 Operating hours

Task	Limit	Operating hours [h]																			
		7,500	7,750	8,000	8,250	8,500	8,750	9,000	9,250	9,500	9,750	10,000	10,250	10,500	10,750	11,000	11,250	11,500	11,750	12,000	12,250
W1096	2 a																				
W0500	-																				
W0501	-																				
W0502	-																				
W0503	-																				
W0505	-																				
W0506	-																				
W0507	-																				
W0508	-																				
W1024	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1009	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1080	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1002	-		X				X					X			X			X			X
W1027	1 a		X				X					X			X			X			X
W1053	1 a		X				X					X			X			X			X
W1001	2 a		X				X					X			X			X			X
W1061	-		X									X						X			X
W1060	-		X									X						X			X
W1174	-		X									X						X			X
W1011	-		X									X						X			X
W1176	2 a		X									X						X			X
W1005	2 a		X									X						X			X
W1189	2 a		X									X						X			X
W1038	-											X									
W2000	-											X									
W2002	-											X									
W2003	-											X									
W2009	-											X									
W2010	-											X									
W2011	-											X									
W2017	-											X									
W2036	-											X									
w = weeks m = months a = years																					

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Task	Limit	Operating hours [h]																			
		7,500	7,750	8,000	8,250	8,500	8,750	9,000	9,250	9,500	9,750	10,000	10,250	10,500	10,750	11,000	11,250	11,500	11,750	12,000	12,250
W2041	-											X									
W2081	-											X									
W2110	-											X									
W2111	-											X									
W2112	-											X									
W2062	-											X									
W1063	-											X									
W3000	18 a																				
W3001	18 a																				
W3003	18 a																				
W3004	18 a																				
W3005	18 a																				
W3009	18 a																				
W3015	18 a																				
W3017	18 a																				
W3018	18 a																				
W3019	18 a																				
W1043	18 a																				
w = weeks m = months a = years																					

15,000 to 22,250 Operating hours

Task	Limit	Operating hours [h]																			
		15,000	15,250	15,500	15,750	16,000	16,250	16,500	16,750	17,000	17,250	17,500	17,750	18,000	18,250	18,500	18,750	19,000	19,250	19,500	19,750
W1096	2 a																				
W0500	-																				
W0501	-																				
W0502	-																				
W0503	-																				
W0505	-																				
W0506	-																				
W0507	-																				
W0508	-																				
W1024	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1009	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1080	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1002	-	X				X				X				X				X			X
W1027	1 a	X				X				X				X				X			X
W1053	1 a	X				X				X				X				X			X
W1001	2 a	X				X				X				X				X			X
W1061	-					X								X							X
W1060	-					X								X							X
W1174	-					X								X							X
W1011	-					X								X							X
W1176	2 a					X								X							X
w = weeks m = months a = years																					

TIM-ID: 0000101764-001